

Math 170E Notes - W4

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1 W4 (Ch2-Random Variables)

Example 1.1. Random Variable Example Two dice are thrown: D_1 & D_2 . Let random variable X be the sum of numbers facing up. Find

1. The pmf of X
2. $E(X)$
3. $M_X(t)$, meaning moment generating function of X .

Solution

Example 1.2. For each of the following, determine the constant c so that $f(x)$ satisfies the conditions of being a pmf for a random variable X , and then depict each pmf as a line graph:

1. $f(x) = x/c, x = 1, 2, 3, 4.$ (A)1 (B)3 (C)4 (D)6 (E)10
2. $f(x) = cx, x = 1, 2, 3, \dots, 10.$ (A) $\frac{1}{3}$ (B) $\frac{1}{10}$ (C) $\frac{1}{45}$ (D) $\frac{1}{55}$ (E) $\frac{1}{100}$
3. $f(x) = c(\frac{1}{4})^x, x = 1, 2, 3, \dots$ (A)3 (B) $\frac{1}{3}$ (C)4 (D) $\frac{1}{4}$ (E)1
4. $f(x) = c(x+1)^2, x = 0, 1, 2, 3.$
5. $f(x) = \frac{c}{(x)(x+1)}, x = 2, 3, 4, \dots$

Hint: Write $f(x) = c/x - c/(x+1)$.

Solution

Example 1.3. PSI-2.2.5 Let the random variable X be the number of days that a certain patient needs to be in the hospital. Suppose X has the pmf

$$f(x) = \frac{5-x}{10}, \quad x = 1, 2, 3, 4$$

If the patient is to receive \$200 from an insurance company for each of the first two days in the hospital and \$100 for each day after the first two days, what is the expected payment for the hospitalization?

Solution